CLASS 8 — MATHS · FORMULA SHEET

Chapter 13: **Algebra Play** | Quick revision, one page

Expressions

- Variable: a letter standing for a number; term: a product of numbers and variables; like terms have the same variable part.
- Add or subtract only like terms: $ax + bx = (a + b)x$.
- Value of an expression: substitute the given value of the variable.

Useful expressions

- Even number: $2n$; odd number: $2n + 1$; consecutive numbers: $n,\ n + 1,\ n + 2$.
- Multiple of k: kn. Two-digit number with digits a, b: $10a + b$; reversed: $10b + a$.
- Three-digit number: $100a + 10b + c$.

Identities

- $(a + b)^2 = a^2 + 2ab + b^2$; $(a - b)^2 = a^2 - 2ab + b^2$; $(a + b)(a - b) = a^2 - b^2$.
- $(n + 1)^2 - n^2 = 2n + 1$; $(a + b)^2 - (a - b)^2 = 4ab$.

Patterns and number tricks

- A rule for a pattern (e.g. matchsticks $3n + 1$): find the difference between successive terms; it is the coefficient of n.
- To prove a trick works for every number, express the steps in a variable and simplify.
- $\overline{ab} + \overline{ba} = 11(a + b)$; $\overline{ab} - \overline{ba} = 9(a - b)$.

MASTERMIND